EUR Doc 047

Appendix A

AMHS-SWIM Gateway Testing Plan	
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AMHS-SWIM Gateway Testing Plan

DOCUMENT CHANGE RECORD

The following table records the history of successive versions of the present document.

Version	Date	Modifications	Affected Sections
0.1	09/04/2025	Initial version for presentation to AST TF/06	All
3.0	06/05/2025	Adopted version (AST TF/06) *Note: The 1st adopted version of this Appendix was assigned the value 3.0 in order to be aligned with the most recent adopted version of the main part of EUR Doc 047*	

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1 INTRODUCTION

1.1 PURPOSE OF THE DOCUMENT

The purpose of the document is to define the functional requirements for AMHS/SWIM Gateway testing procedures.

1.2 DOCUMENT STRUCTURE

Chapter 2 deals with the scope of functions to be tested at SWIM implementations which should be accessible for testing. In addition, the used transport service for SWIM communications is identified and the interface with AMHS at the implemented AMHS/SWIM Gateway is detailed.

Chapter 3 defines the scope of the test environment used for conformance testing. Related test configurations are added with explanations on sequences of exchanged AMHS and SWIM messages. In addition, principles for definitions of test cases are set up including the consideration of the so-called “negative testing.”

Chapter 4 describes the tests of gateway operations converting a user message from AMHS to SWIM.

Chapter 5 describes the tests of gateway operations converting a user message from SWIM to AMHS.

2 TEST SCENARIO

2.1 FUNCTIONALITY OF THE IUT

2.1.1 TECHNICAL SPECIFICATIONS PROVISIONS

The assumption is made that the IUT to be tested provides functionality as specified by the AMHS/SWIM Gateway Specification. Figure 1 identifies the key elements of the AMHS/SWIM Gateway which are addressed by the technical specification.

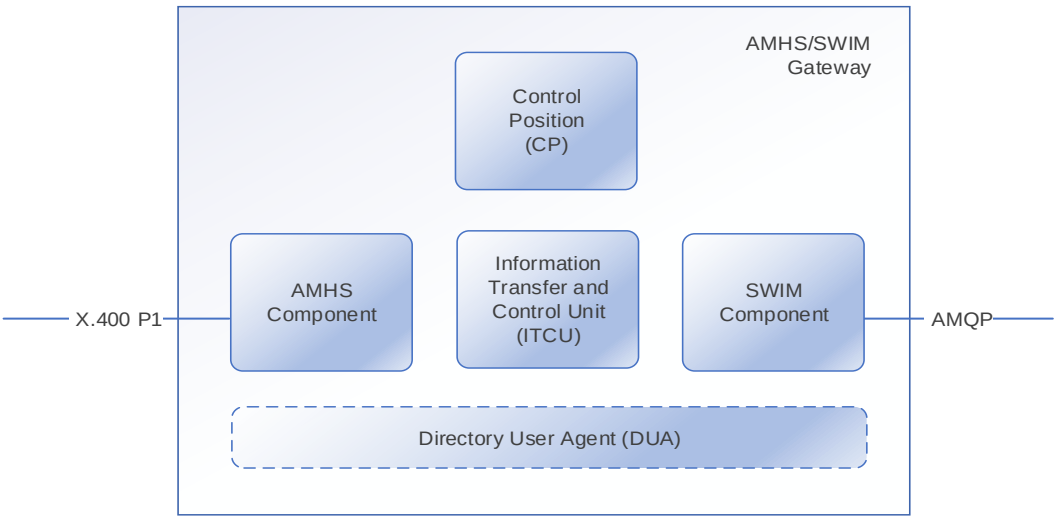

Figure 1: AMHS/SWIM Gateway components

AMHS/SWIM Gateway supports interworking between users of the AMHS and SWIM. The gateway includes an MTA for operations with the transfer level of the AMHS. The mapping functions of the gateway reside in the Information Transfer and Control Unit (ITCU).

Exception handling which needs operator assistance is moved to the Control Position.

The AMHS/SWIM Gateway shall consist of the following functional building blocks:

a) *AMHS Component*: it is responsible for sending messages between the AMHS and the Information Transfer and Control Unit (ITCU)

b) *Information Transfer and Control Unit (ITCU)*: shall convert and transfer information and control the flow of information. It performs bidirectional information mapping and address conversion between AMQP (in the SWIM environment) and X.400 (in AMHS)

c) *SWIM Component*: shall interface with SWIM for exchange of aeronautical information with SWIM information providers and consumers. It provides the interface between the ITCU and SWIM infrastructure using AMQP.

d) *Control Position (CP)*: stores information related with the traffic conveyed by the AMHS/SWIM Gateway as well as the reports on issues occurred during the automated processing that need to be attended by an operator/supervisor.

e) *Directory User Agent (DUA)*: shall interface with the ATN Directory in order to the AMHS address conversion. This component it's considered as optional.

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AMQP is the protocol that has been chosen for the AMHS/SWIM Gateway defined by EUR Doc 047. SWIM Access Point can provide access to the AMHS/SWIM Gateway following any SWIM TI YP compliant interfaces.

2.1.2 IMPLEMENTATION SPECIFIC SWIM FEATURES

Typically, a SWIM implementation includes features which are either beyond the scope of the technical specifications or seen as local issue. For example: User interfaces (HMI) for local submission and delivery.

Such features will not be subject to conformance testing, however, they may be used in support of conformance testing. Example: AMQP message submission at local user interfaces for verification of correct generation of the related AMHS message.

2.2 MODELLING OF THE TEST ENVIRONMENT

The AMHS/SWIM Gateway Implementation Under Test (IUT) is embedded in a simulated operational environment formed by: AMHS Test Tool with two MTA instances and SWIM Test Tool with one Message Broker.

The MTAs have AMHS User Agents (UA) attached, which are used in submission and delivery tests.

It is also possible to make use of the IUT's associated Control Position – if available - to observe outcomes of the conversion process, especially in error situations.

The Test Tool implements two MTA test applications (MTA-1 and -2) to send and receive AMHS messages (IPM, IPN), reports and probes to and from two directions. The test tool generates AMHS data at the X.400/P1/P3/P7 level.

The SWIM Test Tool uses Message Broker to send and receive AMQP messages.

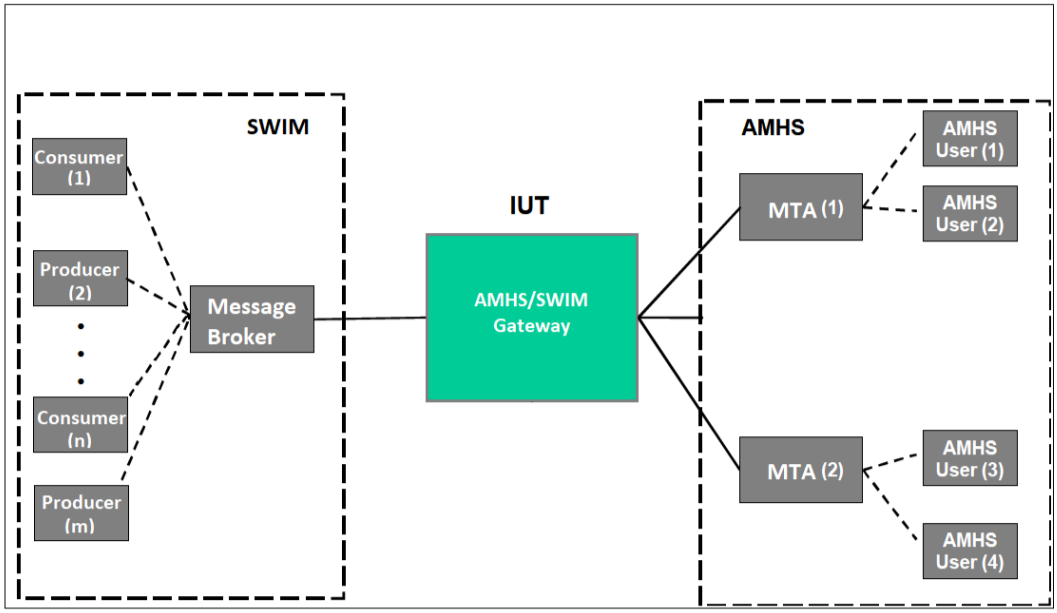

Figure 2 - Operational environment of the IUT

The SWIM Test Tool simulates an operational environment for the IUT as depicted in Figure 2. Figure 3 shows the corresponding model of the test environment. The COM Interfaces of the AMHS Test Tool include two MTA instances representing the two adjacent ATS Message Servers of the IUT and a SWIM Test Tool representing the SWIM environment.

Other major functional components of the SWIM and AMHS Test Tools are the Test Repository (containing predefined test scripts and associated test data), Test Log (containing all the exchanged information objects) and Test Evaluation (performing test evaluations based on the test log against various criteria). The Test Driver controls the execution of test scripts and enters exchanged information objects in the test log. Finally, the component Test Management is tasked with the overall management and administration of the test tool. Additionally, both test tools include a Test Script Editor (providing an HMI for specifications of test scripts and test data).

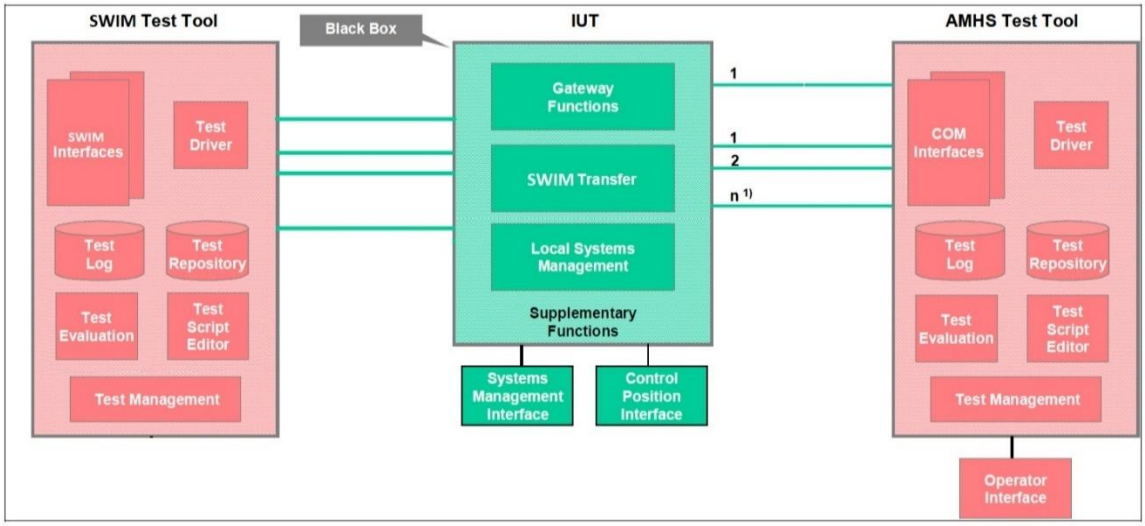

Figure 3 - Model of the test environment

The IUT is represented in the modelled test configuration (Figure 3) by its functional components which will be subject of testing, i.e. SWIM transfer, Gateway functions. The inclusion of other indicated components of the IUT (as Systems Management Interface) will be addressed in subsequent sections.

The two MTA names, the addresses of the n ATS Message User Agents, the SWIM producers and consumers are considered as configuration parameters which are jointly set up in the IUT, SWIM Test Tool and AMHS Test Tool.

Note.– The above outlined functional model includes an abstract, logical view on the SWIM Test Tool for the purpose of understanding the intended testing approach.

2.2.1 WORKING POSITIONS

The scope of conformance testing includes “consumer” and “producer” SWIM components.

Notifications for specified out-of-line situations shall be sent to the Control Position of the AMHS/SWIM Gateway.

Furthermore, traffic logs generated by the IUT during test execution shall be verified against the technical specifications.

The format of input and presentation of test data at the above identified three working positions is seen as a local implementation matter and is, therefore, out of scope of conformance testing. The inclusion of these working positions in testing is limited to observation and interpretation of test data.

Below, the three types of working positions are handled in more detail.

2.2.1.1 AMQP PRODUCERS AND CONSUMERS

The SWIM Test Tool allows the generation of AMQP messages from SWIM to AMHS and in the opposite direction the consumption of AMQP messages that shall be sent to AMQP users and which were originally IPMs.

As part of the SWIM Test Tool, AMQP producers of SWIM information shall generate AMQP messages and queued them in AMQP Message Broker queues. On the other hand, AMQP consumers of SWIM information shall obtain AMQP messages of their own SWIM Message Broker queues.

This interface shall not receive IPNs, Reports or Probes as they are not conveyed in the SWIM environment. These types of AMHS messages shall be reported and logged to Control Position.

The inclusion of consumer and producer operations in conformance testing requires operator observation and interaction during test execution.

2.2.1.2 CONTROL POSITION

There are technical specification requirements to notify the Control Position of the AMHS/SWIM Gateway on specified deviations from the gateway's normal operations.

Notifications may be issued for the operators’ information only or may require operator assistance for recovery from an occurred communication problem. How the Control Position is implemented is out of scope of the technical requirements. However, the correct presentation of notifications at the IUT's Control Position is seen as subject of conformance testing. Appropriate observations have to accompany certain test exercises.

2.2.1.3 ACCESS TO SYSTEMS MANAGEMENT FUNCTIONS

Access to IUT's systems management functions by means of a related working position shall be possible for:

- Preparation of the IUT for the test configuration, and
- Verification of traffic logs generated by the IUT during test exercises.

3 SUPPORTED SCOPE OF CONFORMANCE TESTING

3.1 GENERAL ASPECTS

The scope of testing covers the communications and interoperations between the SWIM and the AMHS by means of the AMHS/SWIM Gateway. The AMHS interface is tested for verification of the gateway's capability to intercommunicate with SWIM, however, it is not subject of dedicated conformance testing.

The following aspects of conformance testing shall be considered:

- Protocol testing, encompassing AMHS to AMQP mappings (and vice versa).
- Functionality testing, to ensure the appropriate implementation of functionalities mentioned on the technical specification

Below, the scope of conformance testing of Gateway Operations will be specified by related test cases.

3.1.1 GATEWAY OPERATIONS

Subject of the Gateway Operations is the bi-directional conversion between AMHS and AMQP user messages and the handling of accompanying service information, i.e., AMQP messages, AMHS IPMs, AMHS Reports, AMHS Probes and AMHS Receipt Notifications.

Considering the functional complexity of the AMHS/SWIM Gateway, the test cases are logically subdivided in two sub-groups:

1) Flow of messages from AMHS to SWIM, including handling of Probes, and

2) Flow of messages from SWIM to AMHS.

The two sub-groups include the handling of accompanying service information.

Regarding AMHS and SWIM, the AMHS/SWIM Gateway acts as an endpoint in exchange of information.

The address look-up tables specified in ICAO Doc 9880 Part II [1] paragraph 4.3.2 for the AFTN/AMHS Gateway shall be used by the AMHS/SWIM Gateway for address conversion.

The IUT's capability to convert addresses in the SWIM and AMHS address spaces is tested in both sub-groups test cases.

The parameters required for testing are defined in EUR Doc 047, AMHS/SWIM Gateway Specification.

The ITCU shall perform long-term logging for a period of at least thirty days of information related to the following exchanges of information objects with the AMHS component and the SWIM component.

Logging requirements are defined in section 4.3.1 Traffic logging of EUR Doc 047, AMHS/SWIM Gateway Specification. The requirements make it possible to perform message tracing when an investigation is needed. The query of the logged information is seen as a local implementation detail.

Verification of the logged information shall be performed on any test case. For this purpose, the local query and tracing provisions will be used. Access is typically provided at the system management interface of the IUT.

3.1.1.1 MESSAGES FROM AMHS TO SWIM

The following actions may be performed with the Gateway test configuration depicted in Figure 4:

- The AMHS Test Tool provides the IUT with valid and invalid AMHS Messages containing IPMs. The IUT converts valid AMHS Messages to AMQP messages, whereas invalid AMHS Messages are rejected by the IUT with Non-Delivery Reports (NDR) which are returned to the AMHS Test Tool.

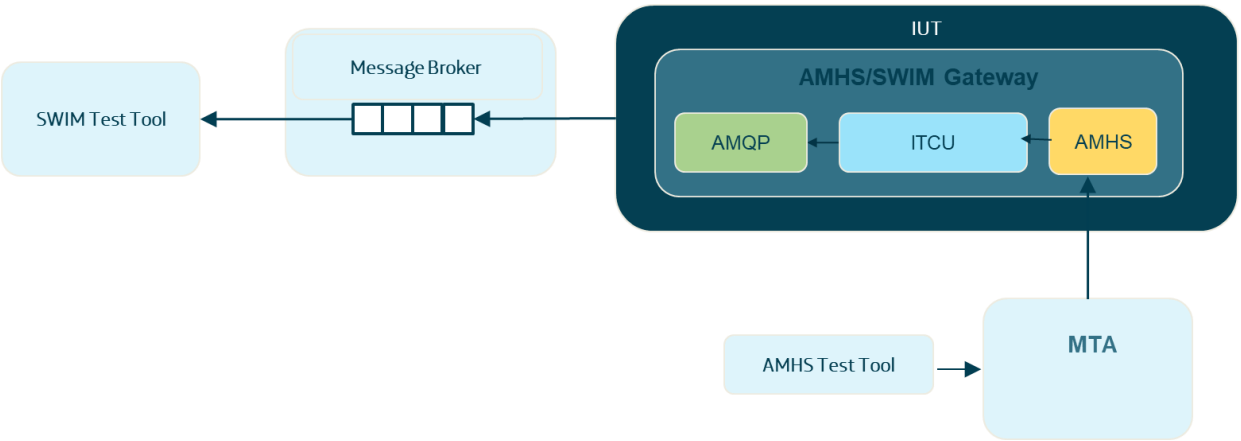

Figure 4 - AMHS to SWIM

For certain out-of-line situations, which may occur during conversion in the AMHS/SWIM Gateway, the technical specification defines error notifications to be forwarded to the Gateway's Control Position. Such notifications must be observed during test execution.

The AMHS/SWIM Gateway does not generate X.400 information objects of class probe.

The following actions may be performed with the Gateway test configuration when receiving a Probe:

- The AMHS Test Tool provides the IUT with valid and invalid Probes, that are logged, and reported to the Control Position but are not conveyed to SWIM.
- The IUT verifies if the probe complies with requirements defined in the technical specification. The IUT generates, depending on the result of the verification, either a Delivery Report (DR) or a Non-Delivery Report (NDR) which is returned to the AMHS Test Tool.

3.1.1.2 MESSAGES FROM SWIM TO AMHS

The following actions may be performed with the Gateway test configuration depicted in Figure 5:

- The SWIM Test Tool provides the IUT with valid and invalid AMQP messages. The IUT converts valid AMQP messages in AMHS messages which leave the IUT, whereas invalid AMQP messages are handled according to locally implemented procedures.

- If the message does not comply with requirements defined in the technical specification, it shall be rejected, logged, and reported to the Control Position.

- The AMHS Test Tool provides the IUT with a Non-Delivery Report (NDR) simulating the detection of an unknown recipient address within the AMHS. The ITCU upon the reception of a AMHS non-delivery report passed from the AMHS component, the report is logged, the situation is reported to the control position; and stored for appropriate processing at the Control Position.

- The SWIM Test Tool provides the IUT with an AMQP message that should be converted to an AMHS message with SS priority. This message should be processed by the AMHS Test Tool generating a Receipt Notification (RN) indicating the reception of a SS-priority message by a specified AMHS recipient. An AMHS RN passed from the AMHS component to the ITCU and not

rejected as the result of 4.4.7.1 or 4.4.7.2 of technical specification, shall be logged and reported to the Control Position.

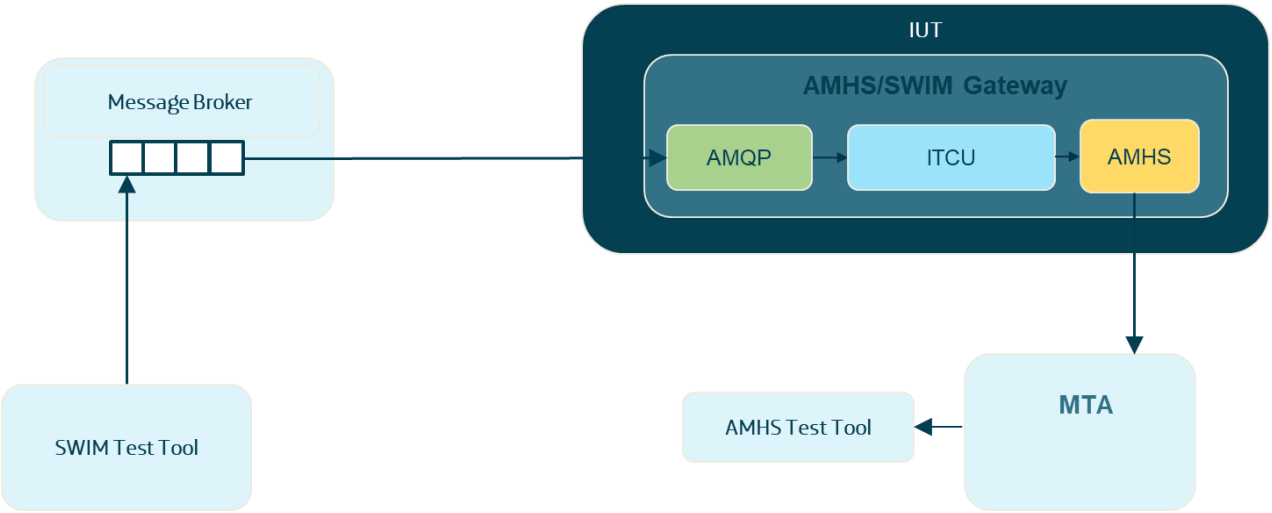

Figure 5 - SWIM to AMHS

For certain out-of-line situations which may occur during conversion in the AMHS/SWIM Gateway, the technical specification defines error notifications to be forwarded to the Gateway's Control Position. Such notifications shall be observed during test execution.

3.2 DEFINITION OF TEST CASES

According to ISO 9646-2 a test case comprises the actions to achieve a specific test purpose.

Each test case normally has a single test purpose, such as that of verifying that the IUT has a certain required capability (i.e., the capability to support certain message lengths) or exhibits a certain required behavior when a particular event occurs.

There may be many criteria for methodical definitions of test cases to achieve an envisaged confidence in a particular functional area of the IUT.

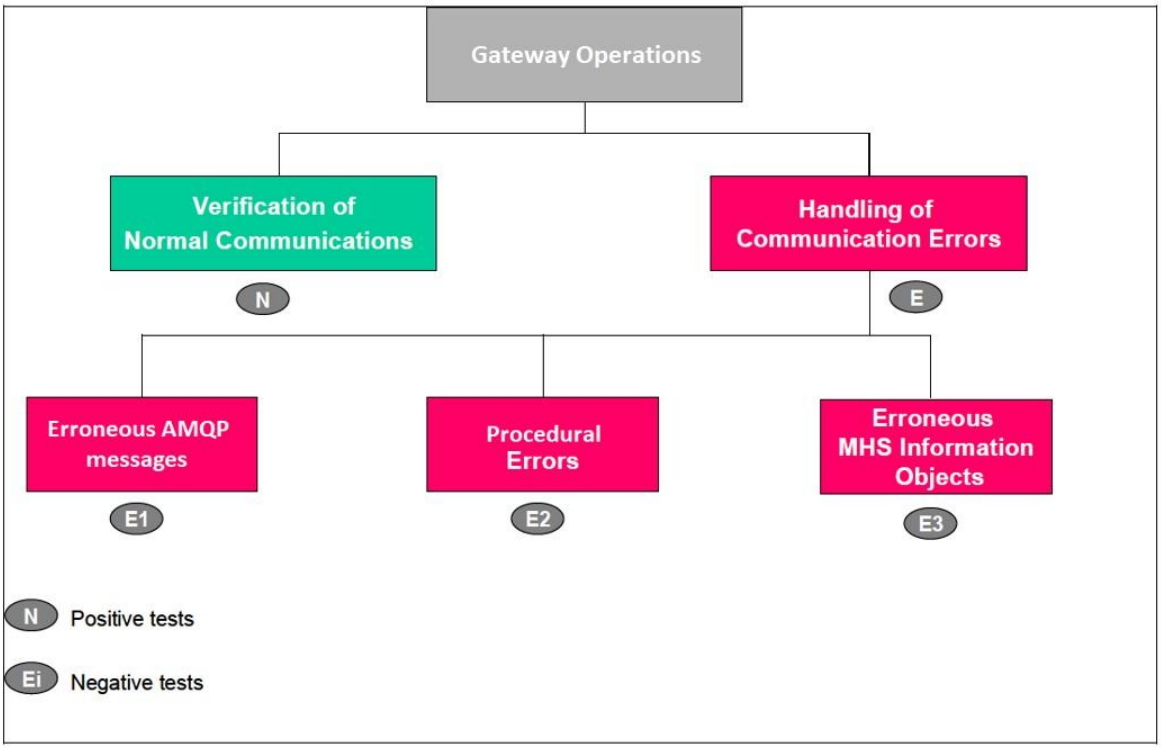

Figure 6 – Test Cases

The test categories are based on the following definitions:

- Normal communications (N). – The Test Tools provide the IUT with correct inputs and the related responses (behaviour) of the IUT should be in compliance with the technical specifications. It should be noted that a correct response of the IUT not always equals to a successful transmission of a SWIM/AMHS Information Object. For example, a valid AMQP Message could be transferred to AMHS and be rejected by a MTA if it can't be routed, causing the reception of an NDR by the IUT. The same applies to AMQP Messages transferred to AMHS and causing RN and NRN.

- Erroneous AMQP messages (E1). – The SWIM Test Tool does not act in compliance with the AMHS/SWIM Gateway Specification. These SWIM conventions are defined in Table 8 – 4.5.2 Use of AMQP elements. It should be noted that violations of such AMQP conventions are not equal with faults in the communication.

- Procedural errors (E2). – The Test Tool does not act in compliance with the procedures, or the arrangements made in the test configuration. Examples: 1) The Test Tool sends a message that have a payload larger than the configured maximum number of bytes. 2) The IUT is requested to send a message with a Report request, however, the AMHS Test Tool does not return any Report.

3) The AMHS Test Tool uses a recipient address which is unknown in the test configuration. 4) The AMHS Test Tool sends a RN or NDR to the AMHS/SWIM Gateway, which is not related to any subject message.

- Erroneous MHS information objects (E3). – That means that arguments have not allowed values or information objects are corrupted. A distinction is made between the levels of IPMS and MTS. Example: Mandatory arguments in the IPM heading are missing or there is no IPM body attached to the IPM heading.

4 GATEWAY OPERATIONS (AMHS TO AMQP)

CTSW001 – CONVERT AN INCOMING IPM TO AMQP FORMAT

CTSW001	**Convert an incoming IPM to AMQP format**
Test criteria	This test is successful, if the IUT converts an IPM into AMQP format correctly.
Scenario description	From the AMHS Test Tool send a sequence of ATS messages (IPMs) over AMHS to the IUT addressing an AMQP consumer. **<u>Basic IPM</u>** The following messages shall contain a basic *ia5-text* body part. • The first ATS message shall have ATS-message-priority KK. • The second ATS message shall have ATS-message-priority GG. • The third ATS message shall have ATS-message-priority FF. • The fourth ATS message shall have ATS-message-priority DD. • The fifth ATS message shall have ATS-message-priority SS. Each message shall have different ATS-filing-time and ATS-message-text and address an AMQP consumer. The optional-heading-information element shall be empty. Verify the messages received at the AMQP interface. Check the correct format of the AMQP message. Verify the priority, both in the AMQP Application properties (*amhs_ats_pri*) and AMQP Header (priority) according to tables 3 and 5 of the EUR Doc 047. Check the filing time (*amhs_ats_ft*) for each received message. The "durable" element of the AMQP header shall be generated as "true". Check that the *amhs_ipm_id* property contains the value of the IPM Identifier. Compare the AMQP message text (*amqp-value*) with the original ATS-message-text. The AMQP element *data* shall be empty. **<u>Extended IPM</u>** The following messages shall contain an extended *general-text-body-part:* • The first message shall have precedence 14. • The second message shall have precedence 28. • The third message shall have precedence 57. • The fourth message shall have precedence 71. • The fifth message shall have precedence 107. • The sixth message shall address two AMQP recipients. One of them shall have precedence 107 and the other, precedence 14.

	Each message shall have different authorization-time and text and address AMQP consumers. The originators-reference element shall be empty, and the subject field shall contain information. Two or more recipients shall contain responsibility element set to “responsible” and at least one shall contain “not responsible”. Verify the messages received at the AMQP interface. Check the correct format of the AMQP message. Verify the priority, both in the AMQP Application properties (*amhs_ats_pri*) and AMQP Header (priority) according to tables 3 and 5 of the EUR Doc 047. For the last message, the value of the priority and the AMQP application property *amhs_ats_pri* shall be mapped and converted from the highest precedence element of the recipients included in the IPM, which is 107. Check that for each of the converted AMQP messages the amhs_ats_ft contains the value of the authorization-time and the amhs_subject the value of the IPM element subject. Check that the *amhs_ipm_id* property contains the value of the IPM Identifier. The “*durable*” element of the AMQP header shall be generated as “*true*”. Compare the AMQP message text (*amqp-value*) with the original ATS-message-text. From the SWIM test tool, check the originator and recipients, both shall be composed of 8-letter addresses. The element amhs_recipients shall be composed of the different entries of the recipient-name in each of the per-recipient-fields that have the responsibility element set to ‘responsible’. In case of multiple recipients, they shall be separated by “,”.
Ref Doc 047	4.4.3 Generation of AMQP message 4.4.3.2.1 Durable 4.4.3.2.2 Priority 4.4.3.3.1 message id 4.4.3.3.5 creation-time 4.4.3.4.1 amhs_ipm_id 4.4.3.4.3 amhs_ats_pri 4.4.3.4.4 amhs_recipients 4.4.3.4.5 amhs_ats_ft 4.4.3.4.7 amhs_originator 4.4.3.4.8 amhs_subject 4.4.3.5.2 amqp-value Table 3. ATS Priority to AMQP Priority conversion Table 5. Amhs_ats_pri ATS-message-priority and IPM precedence equivalency
Test class	Normal communications (N)

CTSW002 – CONVERT AN INCOMING IPM CONTAINING OPTIONAL-HEADING-INFORMATION IN THE ATS-MESSAGE-HEADER OR ORIGINATORS-REFERENCE ELEMENT

CTSW002	**Convert an incoming IPM containing optional-heading-information in the ATS-message-header or originators-reference element**
Test criteria	This test is successful, if the IUT converts an IPM containing optional-heading-information (OHI) in the ATS-message-header or originators-reference element correctly into AMQP format
Scenario description	From the AMHS Test Tool send a sequence of ATS messages (IPMs) over the AMHS to the IUT. The sequence of IPMs shall address a AMQP consumer. • The first ATS message shall have FF priority and contain OHI text. • The second ATS message shall have SS priority and contain OHI text. • The third message shall have precedence value equals to 57 and contain. the originators-reference element. • The fourth message shall have precedence value equals to 107 and contain the originators-reference element. Check the AMQP messages received and verify the AMQP format. Check the format and contents of the OHI application property.
Ref Doc 047	4.4.3.4.6 amhs_ats_ohi
Test class	Normal communications (N)

CTSW003 – GENERATE A DR FOR A SUCCESSFULLY TRANSLATED IPM

CTSW003	**Generate a DR for a successfully translated IPM**
Test criteria	This test is successful, if the IUT returns a DR for a successfully translated ATS message (IPM), if a report was requested by the originator or the originating MTA.
Scenario description	From the AMHS Test Tool send a sequence of ATS messages (IPMs) to the IUT addressing an AMQP consumer. The IPMs shall have ATS-Message-Priority "FF" and different combinations of values for the *originator-report-request* flag and the *originating-MTA-report-request* flag according to the table below. The IUT shall convert all ATS messages into AMQP format and forward the DRs to the AMHS Test Tool if needed. Check the messages received at the AMHS interface, and verify that the IUT sends a DR for every ATS message, if: • the *originator-report-request* element has the abstract value "report", or • the originating-MTA-report-request element has the abstract value "report" or "audited-report", or • both conditions above. (see table below).
Ref Doc 047	4.4.8 Generation of AMHS reports: As specified in ICAO Doc 9880 Part II [1] paragraph 4.5.6: Doc9880_4.5.6.2.20 Generation of AMHS Reports - per-recipient-indicators
Test class	Normal communications (N)

ATS Message	**Value of the originator-report-request element**	**Value of the originating-MTA-report-request element**	**Expected result for conformance test CTSW003**
1	no-report(0)	non-delivery-report(1)	IUT does not return a report
2	no-report(0)	report(2)	IUT returns a DR
3	no-report(0)	audited-report(3)	IUT returns a DR
4	non-delivery-report(1)	non-delivery-report(1)	IUT does not return a report
5	non-delivery-report(1)	report(2)	IUT returns a DR
6	non-delivery-report(1)	audited-report(3)	IUT returns a DR

CTSW004 – GENERATE AN NDR, IF THE ATS-MESSAGE-HEADER HAS A SYNTAX ERROR

CTSW004	**Generate an NDR, if the ATS-message-header has a syntax error**
Test criteria	This test is successful, if the IUT generates an NDR, if it receives an IPM addressed to an AMQP consumer containing erroneous ATS-message-header
Scenario description	From the AMHS Test Tool send a sequence of six messages (IPMs) to the IUT addressed to an AMQP consumer via the IUT's gateway. • The first message (IPM) shall contain an empty ATS-message-priority. • The second message (IPM) shall contain an invalid ATS-message-priority. • The third message (IPM) shall contain an empty ATS-message-filing-time. • The fourth message (IPM) shall contain an invalid ATS-message-filing-time. • The fifth message (IPM) shall contain an empty ATS-message-header and no IHE. Check the messages received at the AMHS and AMQP interfaces of the AMHS Test Tool. Verify that the IUT does not convert the received AMHS messages into AMQP but returns a NDR for each message. Verify that all NDRs contain the following elements (as specified in the Doc 9880, Part II, section 4.5.2.1.5-b): • "unable-to-transfer" for the *non-delivery-reason-code*, • "content-syntax-error" for the *non-delivery-diagnostic-code*, and • "unable to convert to AMQP due to ATS-message-header or Heading Fields syntax error" for the supplementary-information.
Ref Doc 047	4.4.2.5 ATS Message header (Verify message header is present)
Test class	Erroneous MHS information objects (E3)

CTSW005 – GENERATE AN NDR, IF CURRENT TIME EXCEEDS THE LATEST DELIVERY TIME OF IPM

CTSW005	**Generate an NDR, if current time exceeds the latest delivery time of IPM**
Test criteria	This test is successful, if the IUT generates an NDR, if it receives an IPM addressed to an AMQP consumer, and the current time exceeds the value of the *latest-delivery-time* contained in the Message Transfer Envelope (MTE).
Scenario description	From the AMHS Test Tool send two messages (IPMs) to the IUT addressed to an AMQP consumer via the IUT's gateway. • For the first message (IPM), *latest-delivery-time* shall be set to a date in the past. • For the second message (IPM), *latest-delivery-time* shall be later than the current time. Check the messages received at the AMHS and AMQP interfaces of the AMHS Test Tool. Verify that the IUT does not convert the first received AMHS message into AMQP but returns an NDR. Verify that NDR contains the following elements (as specified in the Doc 9880, section 4.5.2.4.11): • "unable-to-transfer" for the *non-delivery-reason-code*, • "maximum-time-expired" for the *non-delivery-diagnostic-code*
Ref Doc 047	4.4.5 Use of message transfer envelope parameters Doc9880_4.5.2.4.11 Latest-delivery-time
Test class	Normal communications (N)

CTSW006 – REJECT AN IPM IF THE SIZE OF THE PAYLOAD IN BYTES EXCEEDS THE MAXIMUM CONFIGURED

CTSW006	**Reject an IPM if the size of the payload in bytes exceeds the maximum configured**
Test criteria	This test is successful, if the IUT rejects a received IPM if the size of the payload in bytes of the AMHS message exceeds the maximum value specified in the AMHS/SWIM Gateway's configuration parameter "Maximum message data size."
Scenario description	From the AMHS Test Tool send three ATS messages (IPMs) to the IUT. The messages shall have the *originator-report-request* and the *originating-MTA-report-request* flag set to "non-delivery-report" and contain the recipient address of an AMQP consumer. a) Send one IPM, with an *ia5-text body part.* The size of the payload shall not exceed the "Maximum message data size" configured. Verify that this message is received at the AMQP test interface. b) Send another IPM, in this case, the *ia5-text* body part size shall exceed the "Maximum message data size" configured. c) Send another IPM, with a *file-transfer-body-part.* The size of the payload shall exceed the "Maximum message data size" configured. Verify that for the last two messages the IUT does not convert the AMHS message into AMQP format, but rejects the message for all the message recipients and returns a NDR with the following elements: • "unable-to-transfer" for the *non-delivery-reason-code*, • "content-too-long" for the *non-delivery-diagnostic-code*, and • "unable to convert to AMQP due to the content size" for the *supplementary-information.*
Ref Doc 047	4.4.2.6 Check of "maximun message data size" 4.4.8 Generation of AMHS reports
Test class	Procedural errors (E2)

CTSW007 – REJECT AN IPM WITH MULTIPLE BODY PART

CTSW007	**Reject an IPM with multiple body parts**
Test criteria	This test is successful, if the IUT generates an NDR, if it receives an IPM addressed to an AMQP consumer containing multiple body parts.
Scenario description	From the AMHS Test Tool send the following ATS messages (IPM) to the IUT. For this test, support is limited to two body parts: one text body part comprising an ATS message header and one FTBP. • The first message shall contain two body parts. One of them shall be basic *ia5-text* and the other one *file-transfer-body-part*. • The second message shall also contain two body parts. One of them shall be *general-text-body-part* and the other one *file-transfer-body-part*. Verify that both messages are received at the AMQP test interface. • The third message containing two ia5-text body parts. • The fourth message shall contain three body parts. One of them shall be a basic *ia5-text*, a second one *general-text-body-part* and the last one *file-transfer-body-part*. Verify that a NDR is generated for the last two messages and received by AMHS Test Tool with the following elements: • "unable-to-transfer" for the *non-delivery-reason-code*, • "content-syntax-error" for the *non-delivery-diagnostic-code*, and • "unable to convert to AMQP due to unsupported body part type" in the third message in the third message and "unable to convert to AMQP due to multiple body parts" in the fourth message for the *supplementary-information*.
Ref Doc 047	4.4.2.2 Verify body parts number 4.4.2.4 Verify two body parts case 4.4.8 Generation of AMHS reports
Test class	Erroneous MHS information objects (E3)

CTSW008 – REJECT AN IPM WITH UNSUPPORTED CONTENT-TYPE

CTSW008	**Reject an IPM with unsupported content-type**
Test criteria	This test is successful, if the IUT's gateway component rejects an incoming message of content-type other than "interpersonal-messaging-1988" and generates an NDR.
Scenario description	From the AMHS Test Tool send a sequence of messages to the IUT addressed to an AMQP consumer. The messages shall have different values for the content-type contained in the MTE. • The 1st message shall contain a *built-in content-type* value "interpersonal-messaging-1988(22)". • The 2nd message shall contain a *built-in content-type* value "interpersonal-messaging-1984(2)". • The 3rd message shall contain a *built-in content-type* value "edi-messaging(35)". • The 4th message shall contain a *built-in content-type* value "unidentified(0)". All messages shall contain an IPM body with ATS-message-header and ATS-message-text. Verify that the IUT accepts and converts the 1st message, but rejects 2nd, the 3rd and 4th message. Verify that the IUT returns a NDR for the 2nd, 3rd and 4th message containing: • "unable-to-transfer" for the *non-delivery-reason-code*, and • "content-type-not-supported" for the *non-delivery-diagnostic-code*.
Ref Doc 047	4.4.1.1 Content-type element 4.4.8 Generation of AMHS reports
Test class	Erroneous MHS information objects (E3)

CTSW009 – DISTRIBUTE AN IPM TO AMHS USERS AND AMQP CONSUMERS

CTSW009	**Distribute an IPM to AMHS users and AMQP consumers**
Test criteria	This test is successful, if the IUT distributes an IPM addressing both AMHS users and AMQP consumers correctly.
Scenario description	From the AMHS Test Tool send two ATS messages (IPMs) addressing both AMHS users and AMQP consumers to the IUT. • The IPM Heading of the first message shall contain two primary recipients, which are one AMHS user and one AMQP consumer and two copy recipients, which are also one AMHS user and one AMQP consumer. All four addresses shall be contained as recipient-names in the Message Transfer Envelope (MTE). • The IPM Heading of the second message shall contain two primary recipients, which are one AMHS user and one AMQP consumer and two blind copy recipients, which are also one AMHS user and one AMQP consumer. Only the two blind-copy addresses shall be contained as recipient-names in the Message Transfer Envelope (MTE). The message shall have the originator-report-request flag set to “non-delivery-report”. Verify that both messages (IPMs) are: • relayed to AMHS, and • relayed and converted to AMQP format. Check the messages received at the AMHS-interface. Verify that: • the first message contains an MTE with all AMHS recipient addresses and an IPM heading with all AMHS and AMQP recipients, and • the second message contains an MTE with the AMHS recipient address, which was the blind-copy recipient and an IPM heading with all AMHS and AMQP recipients. Check the messages received at the AMQP port. Verify that: • the first message contains the addressee indicators of both AMQP consumers, and • the second message contains the addressee indicator of the AMQP consumer, which was the blind-copy recipient.
Ref Doc 047	4.4.3.4.4. BCC and CC Recipients
Test class	Normal communications (N)

CTSW010 – REJECT AN IPM ADDRESSING MORE AMQP CONSUMERS THAN THE MAXIMUM CONFIGURED

CTSW010	**Reject an IPM addressing more AMQP consumers than the maximum configured**
Test criteria	This test is successful, if the IUT rejects a received IPM addressing a number of message recipients, which the ITCU is responsible for conveyance, that exceeds the maximum value defined for the SWIM component in the AMHS/SWIM Gateway's configuration parameter "Maximum message number of recipients."
Scenario description	From the AMHS Test Tool send two ATS messages (IPMs) to the IUT. These messages shall have the *originator-report-request* flag set to "non-delivery-report". a) Send one IPM with 512 recipients, if the "Maximum message number of recipients" parameter is set to 512. Verify that this message is received at the AMQP test interface. b) Send one IPM with 513 recipients, if the "Maximum message number of recipients" parameter is set to 512. Verify that the IUT does not convert the AMHS message into AMQP format, but rejects the message for all the message recipients and returns a NDR with the following elements: • "unable-to-transfer" for the *non-delivery-reason-code*, • "too-many-recipients" for the *non-delivery-diagnostic-code*, and • "unable to convert to AMQP due to number of recipients" for the *supplementary-information.*
Ref Doc 047	4.4.2.7 Check of "maximum message number of recipients" 4.4.8 Generation of AMHS reports
Test class	Procedural errors (E2)

CTSW011 – PROBE CONVEYANCE TEST

CTSW011	**Probe Conveyance Test**
Test criteria	This test is successful, if the IUT (receiving a probe with an AMQP consumer as intended recipient) generates a DR, if conversion to AMQP is possible or an NDR, if conversion to AMQP is not possible.
Scenario description	From the AMHS Test Tool send a sequence of AMHS probes to the IUT. • Probe 1 shall specify a content-length lower than the defined in “Maximum message data size” parameter and address an AMQP consumer reachable via the AMHS/SWIM Gateway. • Probe 2 shall specify a content-length lower than the defined in “Maximum message data size” and address an AMQP consumer, which is routed by the IUT via the gateway (MTCU), but which cannot be mapped onto a valid AMQP address by the MTCU. • Probe 3 shall specify a content-length higher than the defined in “Maximum message data size” and address an AMQP consumer reachable via the AMHS/SWIM Gateway. • Probe 4 shall have a recipient argument addressing 512 AMQP consumers, if the “Maximum message number of recipients” parameter is set to 512. • Probe 5 shall have a recipient argument addressing more than 512 AMQP consumers, if the “Maximum message number of recipients” parameter is set to 512. Check the messages received at the AMHS Test Tool-AMHS interface. Verify that the IUT returns a report for each probe. Check the report contents and determine if it is a DR, NDR or combined report: • A DR shall be returned in response to Probe 1. • A NDR shall be returned in response to Probe 2. • A NDR shall be returned for Probe 3. • A DR shall be returned for Probe 4. • A NDR shall be returned for Probe 5.
Ref Doc 047	4.4.6.2 PROBES (content-length) Maximun message data size 4.4.6.3 PROBES Maximun message number of recipients 4.4.6.6 PROBES - DR (delivery_report) for recipients which were not rejected
Test class	Normal communications (N) Procedural errors (E2)

CTSW012 – REJECT A PROBE FOR UNKNOWN RECIPIENTS BEFORE CONVERSION INTO AMQP FORMAT

CTSW012	**Reject a Probe for unknown recipients before conversion into AMQP format**
Test criteria	This test is successful, if the IUT that receives a Probe addressed to multiple AMQP consumers, rejects the probe for the considered unknown recipients (if their address conversion into an AF-address cannot be achieved)
Scenario description	From the AMHS Test Tool send a Probe to the IUT via AMHS. The message shall have two (primary) recipients addressing two AMQP consumers. Only the AMHS address of the first AMQP consumer can be translated by the MTCU into a valid AMQP addressee indicator, the AMHS address of the second AMQP consumer is unknown and the MTCU cannot find a match in its address look-up table. Verify that the IUT: • returns a NDR for the unknown recipient. • a DR is generated for valid recipient, to indicate the successful result of the probe conveyance test. Verify that the NDR contains the following elements (as specified in the EUR Doc 047, sections 4.4.6.5): • “unable-to-transfer” for the *non-delivery-reason-code*, and • “unrecognised-OR-name” for the *non-delivery-diagnostic-code*
Ref Doc 047	4.4.6.5 PROBES - recipient-names conversion 4.4.6.6 PROBES - DR (delivery_report) for recipients which were not rejected
Test class	Procedural errors (E2)

CTSW013 – REJECT A PROBE WITH UNKNOWN ORIGINATOR ADDRESS

CTSW013	**Reject a Probe with unknown originator address**
Test criteria	This test is successful, if the IUT rejects a Probe if its originator is unknown and the conversion into an AF-address cannot be achieved.
Scenario description	From the AMHS Test Tool send a Probe to the IUT via AMHS. The message shall have an invalid AMHS address in the originator-name element that can be translated by the MTCU as cannot find a match in its address look-up table. This message shall also have two (primary) recipients addressing two AMQP consumers. Verify that the IUT rejects the probe and returns a NDR for all the recipients. Verify that the NDR contains the following elements (as specified in EUR Doc047, section 4.4.6.4): • "unable-to-transfer" for the *non-delivery-reason-code*, and • "invalid-arguments" for the *non-delivery-diagnostic-code,* and • "unable to convert to AMQP due to unrecognized originator O/R address" for the *non-delivery-diagnostic-code*
Ref Doc 047	4.4.6.4 PROBES - originator-name
Test class	Procedural errors (E2)

CTSW014 – INCOMING RN RELATING TO A SUBJECT MESSAGE WITH PRIORITY DIFFERENT FROM "SS"

CTSW014	**Incoming RN relating to a subject message with priority different from "SS"**
Test criteria	This test is successful, if the IUT rejects an RN if the value of the priority indicator of the subject message is different from "SS"
Scenario description	From the AMHS Test Tool send two RNs (with an AMHS user as originator) to the IUT addressing an AMQP consumer. The first RN shall have a subject message related to an IPM with *ATS-message-priority* set to "SS", while the second one will be related to a message with priority indicator set to "DD". Verify that the first AMHS RN is transferred from the AMHS component to the information transfer and control unit and is not rejected. It is also logged and reported to the Control Position. Check that for the second AMHS RN, as the value of the priority indicator was different from "SS", the error has been logged and reported to the Control Position. Check the storage of the RN for appropriate action at the Control Position.
Ref Doc 047	4.4.7.1 a) The subject IPM has/has not been previously generated by the information transfer and control unit 4.4.7.2 RN priority verifications 4.4.7.3 RN valid 4.4.1.2 b) content is an IPN which is a RN
Test class	Normal communications (N)

CTSW015 – INCOMING RN WITHOUT RELATED SUBJECT MESSAGE

CTSW015	Incoming RN without related subject message
Test criteria	This test is successful, if the IUT rejects an RN if the subject message did not pass the Gateway before and reports the situation to the Control Position.
Scenario description	From the AMHS Test Tool send an RN (with an AMHS user as originator) to the IUT addressing an AMQP consumer. The RN shall have a fictitious subject IPM. Check the output of the IUT at the AMHS transfer port and the control position. Verify that the error has been logged and reported to the Control Position. Check the storage of the RN for appropriate action at the Control Position. Verify that a NDR is generated and received by AMHS Test Tool with the following elements: • "unable-to-transfer" for the *non-delivery-reason-code*, • "invalid-arguments" for the *non-delivery-diagnostic-code*, and • "unable to notify RN to SWIM due to misrouted RN" for the *supplementary-information*.
Ref Doc 047	4.4.7.1 NDR Misrouted
Test class	Procedural errors (E2)

CTSW016 – PROCESSING OF THE CURRENT ENCODED-INFORMATION-TYPES (EIT)

CTSW016	**Processing of the current encoded-information-types (EIT)**
Test criteria	This test is successful, if the IUT's gateway component evaluates the current-encoded-information-types contained in the incoming ATS message and: • accepts (and converts) the message, if it contains one of those values specified in section 4.4.2.1 of EUR Doc 047, or • rejects the message if it does not contain any of those values and generates an NDR. These validations also apply to probes. In this case, the IUT's gateway component evaluates the current encoded-information-types and: • accepts and generates a delivery-report, if requested, to indicate the successful result of the probe conveyance test, if the probe contains one of the values specified in section 4.4.1.4 of EUR Doc 047, or • rejects the message if it does not contain any of those values and generates an NDR.
Scenario description	From the AMHS Test Tool send a sequence of ATS messages (IPMs) over AMHS to the IUT addressing an AMQP consumer. The messages shall have the following values for the *current encoded-information-types* (EIT) contained in the Message Transfer Envelope (MTE) • The 1st message shall contain *built-in-encoded-information-types* with value "ia5-text(2)". • The 2nd message shall contain *built-in-encoded-information-types* with value "unknown(0)". • The 3rd message shall contain *extended-encoded-information-types* with OID "2.6.3.4.2" for ia5-text information types. • The 4th message shall contain *extended-encoded-information-types* with OID "2.6.3.4.0" for unknown information types. • The 5th message shall contain *extended-encoded-information-types* with OID {id-cs-eit-authority 1}. • The 6th message shall contain *extended-encoded-information-types* with OID {id-cs-eit-authority 2}. • The 7th message shall contain *extended-encoded-information-types* with OID {id-cs-eit-authority 1} and OID {id-cs-eit-authority 6}. • The 8th message shall contain *extended-encoded-information-types* with OID {id-cs-eit-authority 1}, OID {id-cs-eit-authority 6} and OID {id-cs-eit-authority 100}. • The 9th message shall contain *extended-encoded-information-types* with (invalid) OID {id-cs-eit-authority 3}.

	• The 10th message shall contain *extended-encoded-information-types* with OID {id-cs-eit-authority 1}, OID {id-cs-eit-authority 6} and (invalid) OID {id-cs-eit-authority 7}. • The 11th message shall contain *built-in-encoded-information-types* with value “ia5-text(2)” and *extended-encoded-information-types* with OID “2.6.3.4.2” for ia5-text as well as OID {id-cs-eit-authority 1} and OID {id-cs-eit-authority 6}. • The 12th message shall contain *extended-encoded-information-types* with OID {id-eit-file-transfer 0}. The messages shall contain a body part corresponding to the c*urrent encoded-information-types* value. Verify that all messages with valid EIT argument are accepted by the IUT’s gateway component, converted to AMQP format and received at the AMQP test interface of the AMHS Test Tool. Verify that all messages with any invalid EIT argument are rejected by the IUT and a NDR is returned with the following elements: • “unable-to-transfer” for the *non-delivery-reason-code*, and • “encoded-information-types-unsupported” for the *non-delivery-diagnostic-code*. From the AMHS Test Tool send a sequence of Probes over AMHS to the IUT addressing an AMQP consumer. These probes shall have the previously mentioned values for the *current encoded-information-types* (EIT) contained in the Message Transfer Envelope (MTE). Verify that all the probes with valid EIT argument are accepted by the IUT’s gateway component, a delivery-report is generated, if requested, to indicate the successful result of the probe conveyance test. Verify that all probes with any invalid EIT argument are rejected by the IUT and a NDR is generated.
Ref Doc 047	4.4.2.1 Verify encoded information types 4.4.6.1 PROBES Verify encoded information types 4.4.1.4 PROBES abstract-value of the content-type element
Test class	Erroneous MHS information objects (E3)

CTSW017 – INCOMING IPM WITH AN IA5-TEXT-BODY-PART

CTSW017	**Incoming IPM with an ia5-text-body-part**
Test criteria	This test is successful, if the IUT's gateway component accepts a received ATS message (IPM) with an ia5-text-body-part and converts the IPM into AMQP format correctly.
Scenario description	From the AMHS Test Tool send a sequence of ATS messages (IPMs) over AMHS to the IUT addressing an AMQP consumer. • The first message shall contain an *ia5-text-body-part*, which includes an ATS-message-header and ATS-message-text with IA-5 characters. The *original-encoded-information-types* attribute shall contain *extended-encoded-information-types* with OID "2.6.3.4.2" (ia5-text). • The second message shall be equal except for the *original-encoded-information-types*, which has a built-in value for ia5-text(2) [1] • The third message shall be equal to the first, but the *repertoire* argument in the body shall be ita2(2)) [2]. Verify that the first and the second message are accepted by the IUT's gateway component, converted to AMQP format according to the Table 6 of the EUR Doc 047 and received at the AMQP test interface. Check whether the third message is converted into AMQP format or rejected by the IUT. In case of rejection verify that an NDR is returned with the following elements: • "unable-to-transfer" for the *non-delivery-reason-code*, • "content-syntax-error" for the *non-delivery-diagnostic-code*, and • "unable to convert to AMQP due to unsupported body part type" for the *supplementary-information*.
Ref Doc 047	4.4.2.3 Verify body part type - ia5-text / ia5-text-body-part 4.4.3.3.3 Content-type 4.4.3.4.9 amhs_body_part_type, amhs_content_encoding Table 6. AMHS body part types translation to SWIM
Test class	Erroneous MHS information objects (E3)

Note 1. – It is assumed that an extended ia5-text-body-part can be associated with either a built-in EIT or extended EIT value for ia5-text

Note 2.- The X.400 argument repertoire is based on an enumeration type and takes one of the values ita2 (2) or ia5 (5). In absence of the argument, the default value ia5 is assumed.

CTSW018 – INCOMING IPM WITH A GENERAL-TEXT-BODY-PART AND ISO 646 REPERTOIRE

CTSW018	**Incoming IPM with a general-text-body-part and ISO 646 repertoire**
Test criteria	This test is successful, if the IUT's gateway component accepts a received ATS message (IPM) with a general-text-body-part of which the repertoire set description is Basic (ISO 646) and converts the IPM into AMQP format correctly.
Scenario description	From the AMHS Test Tool send a sequence of ATS messages (IPMs) over AMHS to the IUT addressing an AMQP consumer. All messages shall contain a general-text-body-part, which includes an *ATS-message-header* and *ATS-message-text* . The *original-encoded-information-types* shall be set to *extended-encoded-information-types* with OID {id-cs-eit-authority 1} and OID {id-cs-eit-authority 6} and the *conversion-with-loss-prohibited* argument shall be set to "*conversion-with-loss-allowed*". • The 1st message shall contain character set registration numbers 1 and 6, which specify the Basic ISO 646 repertoire, and the text shall contain ISO 646 (US-ASCII) characters only. • The 2nd message shall contain character set registration numbers 1 and 6, which specify the Basic ISO 646 repertoire, but the text shall contain US-ASCII as well as non-listed US-ASCII characters. Verify that both messages are accepted by the IUT's gateway component, converted to AMQP format and received at the AMQP interface of the AMHS Test Tool. Analyze the received AMQP messages with respect to the AMQP message text.
Ref Doc 047	4.4.2.3 Verify body part type - general-text-body-part_ISO646 / ats-basic-general-text-body-part_ISO646 4.4.3.3.3 Content-type 4.4.3.4.9 amhs_body_part_type, amhs_content_encoding Table 6. AMHS body part types translation to SWIM 4.4.4.5 General-text-body-part
Test class	Normal communications (N)

CTSW019 – INCOMING IPM WITH A GENERAL-TEXT-BODY-PART AND A REPERTOIRE DIFFERENT FROM ISO 646

CTSW019	**Incoming IPM with a general-text-body-part and a repertoire different from ISO 646**
Test criteria	This test is successful, if the IUT's gateway component processes a received ATS message (IPM) with a general-text-body-part of which the repertoire set description is different from ISO 646 according to its local AMHS Management Domain policy[1].
Scenario description	From the AMHS Test Tool send a sequence of ATS messages (IPMs) over AMHS to the IUT addressing an AMQP consumer. All messages shall contain a general-text-body-part, which includes an ATS-message-header and ATS-message-text. The *original-encoded-information-types* shall be set to *extended-encoded-information-types* with eit-authority 100} and the *implicit-conversion-prohibited* argument shall be set to "implicit-conversion-allowed". The message text (data part) shall include ISO 646 (US-ASCII) and non-listed US-ASCII characters. The parameter argument in the IPM body part shall specify the following character sets: • The 1st message shall contain character set registration numbers 1, 6 and 100 which specify the ISO 8859-1 repertoire, i.e. ASCII plus Western European supplementary set • The 2nd message shall contain character set registration numbers, which specify a repertoire from one of the following listed in A.1.3.2 of ISO/IEC ISP 12062-2: Cyrillic, Arabic, Greek or Hebrew. • The 3rd message shall contain character set registration numbers, which specify a repertoire from one of the following listed in A.1.3.2 of ISO/IEC ISP 12062-2: Chinese, Japanese or Korean. Check if the messages are converted or rejected by the IUT according to its local policy. In case of conversion, analyze the received AMQP messages with respect to the characters contained in the AMQP message text. In case of message rejection, check, if the NDR returned contains the following elements: • "unable-to-transfer" for the *non-delivery-reason-code*, • "content-syntax-error" for the *non-delivery-diagnostic-code*, and • "unable to convert to AMQP due to unsupported encoded-information-types" for the *supplementary-information*.
Ref Doc 047	4.4.2.3 Verify body part type - general-text-body-part_ISO8859-1 / ats-basic-general-text-body-part_ISO8859-1 / general-text-body-part_ISO8859-2 4.4.3.3.3 - Content-type 4.4.3.4.9 - amhs_body_part_type, amhs_content_encoding

	Table 6. AMHS body part types translation to SWIM 4.4.4.5 General-text-body-part
Test class	Erroneous MHS information objects (E3)

Note 1.– Depending on the local policy of the AMHS Management Domain a received message with a general-text-body-part of which the repertoire set description is different from ISO 646 can be converted or rejected.

CTSW020 – NOTIFY SS TO CONTROL POSITION

CTSW020	**Notify SS to Control Position**
Test criteria	This test is successful, if the IUT logs and reports to the Control Position when the responsibility element of the corresponding per-recipient-fields of the Message Transfer Envelope (MTE) has the value "*responsible*" and: • the highest *precedence* element of the recipient-extensions included in the IPM has the value 107 (extended IPM), or • the *priority-indicator* of a received AMHS message has the value "SS" (basic IPM) Also, the IUT shall convert the mentioned IPM into AMQP format correctly.
Scenario description	From the AMHS Test Tool send a sequence of messages (IPMs) to the IUT. • The first message (IPM) shall address two AMQP consumers reachable via the IUT's gateway. The *responsibility* element for both shall be "*responsible*". On the other hand, the *precedence* for the first AMQP consumer shall be 107 and, for the other one, shall be 28. • The second message (IPM) shall address two AMQP consumers with responsibility element set to "*responsible*" and contain *ATS-message-priority* set to "SS". • The third message (IPM) shall address an AMQP consumer with responsibility element set to "*responsible"* and an AMHS user with responsibility element set to "*non-responsible"*. The precedence for both shall be 107. • The fourth message (IPM) shall address two AMQP consumers reachable via the IUT's gateway. The *responsibility* element for both shall be "*responsible*" and *precedence* shall be 14. • The fifth message (IPM) shall address two AMQP consumers with responsibility element set to "*responsible*" and contain *ATS-message-priority* set to "DD". Check the messages received at the AMQP interface. Check the correct format of the AMQP message. Verify that, except for the last two messages, the situation has been logged and reported to the Control Position for appropriate action.
Ref Doc 047	4.4.4.4 - Notify SS or precedence 107 to Control position
Test class	Normal communications (N)

5 GATEWAY OPERATIONS (AMQP TO AMHS)

CTSW101 – CONVERT AN AMHS UNAWARE AMQP MESSAGE TO AMHS

CTSW101	**Convert an AMHS unaware AMQP message to AMHS**
Test criteria	This test is successful, if the IUT converts an AMQP message with minimal required (AMHS-unaware service level) information to an AMHS message (IPM) correctly. Check that parameters related to AMHS-unaware service level are correct. *Note.— The AMHS-unaware service level is intended for SWIM information providers and consumers not being aware of the exchange of aeronautical information with AMHS users by means of the AMHS/SWIM Gateway.* Check value of "ATSMHS Service Level" parameter in order to determine whether Basic or Extended should be generated. *Note.— This parameter is not under testing in this specific test case. It will be tested on CTSW103*
Scenario description	From the AMQP test interface send an AMQP text message to the IUT. The AMQP text message shall address a remote AMHS user and consist of AMQP priority property with default value (4) and specify a text payload within the AMQP property amqp-value. Check that the AMHS Test Tool receives an AMHS message from the IUT via the AMHS transfer port. In particular, verify that: • the payload of the received AMHS message contains the text send from the AMQP test interface. • the AMQP default priority has been correctly translated to the element *ATS-message-priority or the IPM precedence and to the* AMHS message transfer envelope *priority* depending on "ATSMHS Service Level" parameter. • the filing time corresponding with the creation time of the AMQP message is correctly inserted and converted to format DDhhmm in the *ATS-message-header or* in the A*uthorization-time* if the message contains IHE depending on "ATSMHS Service Level" parameter. • the addressee indicator of the application property amhs_recipients is correctly translated in the corresponding AMHS OR address and entered as *primary-recipient* in the IPM heading and as *recipient-name* in the MTE. • the trace-information or the element internal-trace information element are in accordance with sections 4.5.4.27-4.5.4.31 of EUR Doc 047. • the sub-components *country-name, administration-domain-name* and *private-domain-identifier* of the component *global-domain-identifier* of the element *message-identifier* correspond to the values of the respective AMHS management domain operating the AMHS/SWIM gateway and that the component *local-identifier* is different from the local-identifier of any other AMHS message, probe or report generated in the AMHS management domain operating the AMHS/SWIM Gateway as specified in section 4.5.4.17 of EUR Doc 047.

	The following components of the element *per-message-indicators* are present in the MTE: • *alternate-recipient-allowed* with the abstract value “alternate-recipient-allowed” • *disclosure-of-other-recipients* takes the abstract value “disclosure-of-other-recipients-prohibited” • *implicit-conversion-prohibited* takes the abstract value “implicit-conversion-allowed” • *content-return-request* takes the abstract value “content-return-not-requested” The following components of the element *extensions* are not present or are present in the MTE with their default values: • *recipient-reassignment-prohibited* with the abstract value “recipient-reassignment-allowed” • *dl-expansion-prohibited* takes the abstract value “dl-expansion-allowed” • *conversion-with-loss-prohibited* takes the abstract value “conversion-with-loss-allowed” The components of the element *per-recipient-indicators* are generated taking the following abstract-values: • “responsible” for the *responsibility* element • “non-delivery-report” for the *originating-MTA-report-request* element • “non-delivery-report” for the *originator-report-request* element From the AMQP test interface send a AMQP binary message. The AMQP message shall address a remote AMHS user and consist of AMQP priority property with default value (4) and specify binary payload within AMQP data property. Check the IPMs that the AMHS Test Tool receives an AMHS message from the IUT via the AMHS transfer port. In particular, verify that: • the payload of the received AMHS message contains the binary content send from the AMQP test interface. • the AMQP default priority has been correctly translated to the element *ATS-message-priority or the IPM precedence and to the* AMHS message transfer envelope *priority* depending on “ATSMHS Service Level” parameter. • the filing time corresponding with the creation time of the AMQP message is correctly inserted and converted to format DDhhmm in the *ATS-message-header or* in the A*uthorization-time* if the message contains IHE depending on “ATSMHS Service Level” parameter. • the addressee indicator of the application property amhs_recipients is correctly translated in the corresponding AMHS OR address and entered as *primary-recipient* in the IPM heading and as *recipient-name* in the MTE.

	• the trace-information or the element internal-trace information element are in accordance with sections 4.5.4.27-4.5.4.31 of EUR Doc 0474.. • the sub-components *country-name*, *administration-domain-name* and *private-domain-identifier* of the component *global-domain-identifier* of the element *message-identifier* correspond to the values of the respective AMHS management domain operating the AMHS/SWIM gateway and that the component *local-identifier* is different from the local-identifier of any other AMHS message, probe or report generated in the AMHS management domain operating the AMHS/SWIM Gateway as specified in section 4.5.4.17 of EUR Doc 047. The following components of the element *per-message-indicators* are present in the MTE: • *alternate-recipient-allowed* with the abstract value "alternate-recipient-allowed" • *disclosure-of-other-recipients* takes the abstract value "disclosure-of-other-recipients-prohibited" • *implicit-conversion-prohibited* takes the abstract value "implicit-conversion-allowed" • *content-return-request* takes the abstract value "content-return-not-requested" The following components of the element *extensions* are not present or are present in the MTE with their default values: • *recipient-reassignment-prohibited* with the abstract value "recipient-reassignment-allowed" • *dl-expansion-prohibited* takes the abstract value "dl-expansion-allowed" • *conversion-with-loss-prohibited* takes the abstract value "conversion-with-loss-allowed" The components of the element *per-recipient-indicators* are generated taking the following abstract-values: • "responsible" for the *responsibility* element • "non-delivery-report" for the *originating-MTA-report-request* element • "non-delivery-report" for the *originator-report-request* element
Ref Doc 047	4.5.2.2 amhs_ats_pri 4.5.2.9 amhs_recipients 4.5.3.3 primary-recipients 4.5.4.12 recipient-reassignment-prohibited, dl-expansion-prohibited and conversion-with- loss-prohibited implies, are assumed to take their default abstract-values 4.5.4.13 recipient-name 4.5.4.15 per-recipient-indicators 4.5.4.16 global-domain-identifier 4.5.4.17 local-identifier 4.5.4.18 country-name 4.5.4.19 administration-domain-name

	4.5.4.20 private-domain-identifier 4.5.4.21 disclosure-of-other-recipients 4.5.4.22 implicit-conversion-prohibited 4.5.4.23 alternate-recipient-allowed 4.5.4.24 content-return-request 4.5.4.25 AMHS management domain 4.5.4.27 global-domain-identifier in the trace-information or in the internal-trace-information 4.5.4.28 arrival-time in the first element of trace-information or of internal-trace-information 4.5.4.29 routing-action 4.5.4.30 mta-name 4.5.4.31 4.5.1.9
Test class	Normal communications (N)

CTSW102 – REJECT AN AMQP MESSAGE IF IT DOES NOT CONTAIN THE MINIMUM INFORMATION NECESSARY TO TRANSFER TO AMHS (AMHS UNAWARE)

CTSW102	**Reject an AMQP message if it does not contain the minimum information necessary to transfer to AMHS (AMHS unaware)**
Test criteria	This test is successful, if the IUT rejects a received AMQP message if it does not contain the minimum information necessary to transfer. For AMHS unaware service level, the minimum information necessary to transfer SWIM Information Objects to AMHS is: • *priority* element in header section. • message-id element in properties section. • creation-time element in properties section. • *data* or *amqp-value* element in application data section. • *amhs_recipients* in application properties section. • *content-type* element in properties section. Note that some of these fields are generally set by the AMQP APIs and can't be changed, so generation of messages with these fields is specified as optional in scenario description.
Scenario description	From the SWIM Test Tool send a sequence of AMQP messages to the IUT addressing an AMHS user. **<u>Text messages</u>** The following messages shall contain <*text/plain; charset="utf-8"*> content-type. • The first AMQP message shall have 10 as *priority*. • Optionally, the second AMQP message shall have empty *message-id* element. • Optionally, the third AMQP message shall have 0 as *creation-time*. • The fourth AMQP message shall have empty *amqp-value*. • The fifth AMQP message shall have empty *amhs_recipients* element. Optionally, another AMQP message shall have empty content-type element. **<u>Binary messages</u>** The following messages shall contain <*application/octet-stream*> content-type. • The first AMQP message shall have 10 as *priority*. • Optionally, the second AMQP message shall have empty *message-id* element. • Optionally, the third AMQP message shall have 0 as *creation-time*.

	• The fourth AMQP message shall have empty *data* element. • The fifth AMQP message shall have an address of more than eight letters as amhs_recipients element. Optionally, another AMQP message shall have empty content-type element. Verify that all messages are rejected, logged, and reported to the Control Position.
Ref Doc 047	4.5.1.1 priority 4.5.1.2 message-id 4.5.1.3 creation-time 4.5.1.4 amqp-value 4.5.1.5 amhs_recipients 4.5.2.2 correct value of priority 4.5.2.10 correct value of creation-time
Test class	Erroneous AMQP messages (E1)

CTSW103 – CONVERSION TO IPM ACCORDING TO ATSMHS SERVICE LEVEL PARAMETER

CTSW103	**Conversion to IPM according to ATSMHS Service Level parameter**
Test criteria	This test is successful, if the IUT converts any AMQP messages to AMHS messages (IPM) correctly according to the "ATSMHS Service Level" parameter. Since implementation of all the possible values of the ATSMHS Service Level parameter is not mandatory, this test must be adjusted to test only the implemented options.
Scenario description	From the AMQP test interface send a sequence of AMQP messages to the IUT. The sequence of AMQP messages shall address a remote AMHS user and consist of seven messages. **<u>ATSMHS Service Level – Basic</u>** For the first message, the ATSMHS Service Level parameter shall be set to "basic". Then AMQP elements susceptible to being translated using extended or basic ATSMHS service level will be mapped to basic service level. Check the IPM that the AMHS Test Tool receives. Verify that the IUT has converted the message correctly accordingly: • *amhs_ats_ohi* shall be conveyed as the value of the optional heading information element in the *ATS-message-Optional-Heading-Info* • *amhs_ats_ft* shall be conveyed as the value of the *ATS-message-Filing-Time* • *amhs_ats_pri* shall be conveyed as the value of the priority-indicator of the *ATS-message-priority* element of the AMHS message, in compliance with the mapping specified in the Table 9 of the EUR Doc 047 • *amqp-value* shall be conveyed as the value of the *ATS-message-text* of the AMHS message The second AMQP message shall contain binary content. In this case, the message will be rejected. Check that the situation has been logged and reported to the Control Position for appropriate action. **<u>ATSMHS Service Level – Extended.</u>** For the third message, the ATSMHS Service Level parameter shall be set to "extended". The AMQP elements susceptible to being translated using extended or basic ATSMHS service level will be mapped to extended service level. Check the IPM that the AMHS Test Tool receives. Verify that the IUT has converted the message correctly according:

	• *amhs_ats_ohi* shall be conveyed as the value of the *originators-reference* of the IPM heading. • *amhs_ats_ft* shall be conveyed as the value of the element *Authorization-time* if the message contains IHE and converted to format: DDhhmm. • *amhs_ats_pri* shall be mapped into one of the authorized values of the precedence element of the recipient extensions of the recipient-specifier as specified in the Table 9 of the EUR Doc 047. • the value of *precedence-policy-identifier* shall be 1.3.27.8.0.0 **<u>ATSMHS Service Level – Content based.</u>** For the fourth message, the ATSMHS Service Level parameter shall be set to "content-based" and *content-type* of the AMQP message shall be *"<application/octet-stream>"* Check the IPM that the AMHS Test Tool receives. Verify that the IUT has converted the messages correctly according to the test *"ATSMHS Service Level – Extended."* For the fifth message, the ATSMHS Service Level parameter shall be set to "content-based" and *content-type* of the AMQP message shall be *"<text/plain; charset="utf-8">"* Check the IPM that the AMHS Test Tool receives. Verify that the IUT has converted the messages correctly according to the test *"ATSMHS Service Level – Basic."* **<u>ATSMHS Service Level – Recipient based</u>** For the sixth message, ATSMHS Service Level parameter shall be set to "recipient-based" and all the recipients of the converted AMHS message shall support the extended format. Check the IPM that the AMHS Test Tool receives. Verify that the IUT has converted the messages correctly according to the test *"ATSMHS Service Level – Extended."* For the seventh message, ATSMHS Service Level parameter shall be set to "recipient-based" and all the recipients, except for one, of the converted AMHS message shall support the extended format. Check the IPM that the AMHS Test Tool receives. Verify that the IUT has converted the messages correctly according to the test *"ATSMHS Service Level – Basic."*
Ref Doc 047	4.5.3.7 IPM heading fields - using extended ATSMHS service level 4.5.3.8 precedence-policy-identifier 4.5.3.9 ATS-Message-Header - using basic ATSMHS service leve3.3.3 ATSMHS Service Level
Test class	Normal communications (N)

CTSW104 – CONVERT INCOMING AMQP MESSAGES WITH DIFFERENT PRIORITIES TO AMHS

CTSW104	**Convert incoming AMQP messages with different priorities to AMHS**
Test criteria	This test is successful, if the IUT converts an AMQP message with minimal required information to an AMHS message (IPM) correctly.
Scenario description	From the AMQP test interface send a sequence of nine AMQP messages to the IUT. The sequence of AMQP messages shall address a remote AMHS user and consist of ten messages, one for each AMQP priority, i.e., according to the Table 9 of the EUR Doc 047 and specifying a text payload within the AMQP application data amqp-value. Check the IPMs that the AMHS Test Tool receives from the IUT via the AMHS transfer port. Verify that the IUT has converted the priority of the messages correctly according to Table 9 of the EUR Doc 047. From the AMQP test interface send a sequence of five AMQP messages to the IUT. The sequence of AMQP messages shall address a remote AMHS user and consist of five messages, containing default AMQP priority (4), and one of each of the possible values of the AMQP application property amhs_ats_pri (SS,DD,FF,GG,KK). Check the IPMs that the AMHS Test Tool receives from the IUT via the AMHS transfer port. Verify that the IUT has converted messages priority correctly according to Table 9 of the EUR Doc 047. From the AMQP test interface send a sequence of four AMQP messages to the IUT. The sequence of AMQP messages shall address a remote AMHS user and consist of four messages, containing AMQP priority 1, and one of each of the possible values of the AMQP application property amhs_ats_pri (SS,DD,FF,GG). From the AMQP test interface send an AMQP message to the IUT. The AMQP message shall address a remote AMHS user and consist of one message, containing default AMQP priority 9 and AMQP application property amhs_ats_pri with value KK. Check the IPMs that the AMHS Test Tool receives from the IUT via the AMHS transfer port. Verify that the IUT has considered that AMQP application property amhs_ats_pri takes precedence over AMQP priority value. That is, verify that the IUT has converted AMQP application property amhs_ats_pri according to Table 9 of the EUR Doc 047.
Ref Doc 047	4.5.2.2 amhs_ats_pri Table 9. Mapping of AMQP priority
Test class	Normal communications (N)

CTSW105 – CONVERT INCOMING AMHS AWARE AMQP MESSAGES CONTAINING FILING TIME TO AMHS

CTSW105	**Convert incoming AMHS aware AMQP messages containing filing time to AMHS**
Test criteria	This test is successful, if the IUT converts an AMQP message with minimal required information to an AMHS message (IPM) correctly.
Scenario description	From the AMQP test interface send a sequence of AMQP messages to the IUT. The sequence of AMQP messages shall address a remote AMHS user and consist of: • The 1st AMQP message shall have empty *amhs_ats_ft* application property. • The 2nd AMQP message shall have “250102” as *amhs_ats_ft* application property. Check the IPMs that the AMHS Test Tool receives from the IUT via the AMHS transfer port. Verify that: • For the first message, the filing time corresponding to the creation time of the AMQP message is correctly inserted and converted to DDhhmm format in the *ATS-message-header or* in the *Authorization-time* depending on “ATSMHS Service Level” parameter, as specified in EUR Doc 047 – 4.5.2.10.1.b • For the second message, the filing time from amhs_ats_ft application property shall be conveyed as the value of the ATS-message-Filing-Time in the ATS-message-header *or* in the *Authorization-time* depending on “ATSMHS Service Level” parameter as specified in EUR Doc 047 – 4.5.2.10.1.b
Ref Doc 047	4.5.2.10 ATS-message-Filing-Time
Test class	Normal communications (N)

CTSW106 – CONVERT AN AMHS AWARE AMQP MESSAGE CONTAINING OPTIONAL HEADING INFORMATION

CTSW106	**Convert an AMHS aware AMQP message containing optional heading information**
Test criteria	This test is successful, if the IUT converts an AMQP message containing optional heading information (OHI) correctly into an AMHS message (IPM).
Scenario description	From the AMQP test interface send a sequence of AMQP messages to the IUT. The sequence of AMQP messages shall address a remote AMHS user and consist of: • The 1st AMQP message shall have “4” as priority and contain amhs_ats_ohi text of less than 53 characters. • The 2nd AMQP message shall have “4” as priority and contain amhs_ats_ohi text of exactly 53 characters. • The 3rd AMQP message shall have “4” as priority and contain amhs_ats_ohi text of more than 53 characters. • The 4th AMQP message shall have “6” as priority and contain amhs_ats_ohi text of less than 48 characters. • The 5th AMQP message shall have “6” as priority and contain amhs_ats_ohi text of exactly 48 characters. • The 6th AMQP message shall have “6” as priority and contain amhs_ats_ohi text of more than 48 characters. Verify that the IUT has converted the messages correctly to extended AMHS messages with appropriate originators-reference element. Check that for the 3rd and 6th messages, amhs_ats_ohi text of AMHS message has been trimmed down, to 53 for the case of the 3rd message and to 48 for the 6th message. Optionally, if Basic ATSMHS service level is implemented by the IUT, then repeat this sequence forcing the generation of ATS-Message-Header and verifying that the IUT has conveyed appropriate value of amhs_ats_ohi text to the ATS-Message-Header.
Ref Doc 047	4.5.2.11 amhs_ats_ohi
Test class	Normal communications (N)

CTSW107 – CONVERT AN AMQP MESSAGE CONTAINING SUBJECT

CTSW107	**Convert an AMQP message containing subject**
Test criteria	This test is successful, if the IUT converts an AMQP message subject correctly into an AMHS message (IPM).
Scenario description	From the AMQP test interface send a sequence of AMQP messages to the IUT. The sequence of AMQP messages shall address a remote AMHS user and consist of: • The 1st AMQP message shall include the AMQP subject property with size value exceeding 128 characters and empty AMQP amhs_subject application property • The 2nd AMQP message shall include the AMQP subject property and empty AMQP amhs_subject application property. • The 3rd AMQP message shall have empty AMQP subject property and “Subject example” as AMQP amhs_subject application property. • The 4th AMQP message shall have “subject from properties” as AMQP subject property and “subject from application property field” as AMQP amhs_subject application property. Verify that the IUT has converted the messages correctly. Check that for the 1st AMHS message the property subject was trimmed to 128 characters and mapped to the IPM element subject. Check that for the 2nd AMHS message the property subject was mapped to the IPM element subject. Check that for 3rd and 4th message, the application property amhs_subject was mapped to the IPM element subject.
Ref Doc 047	4.5.2.3 amhs_subject
Test class	Normal communications (N)

CTSW108 – INCOMING AMQP MESSAGE WITH KNOWN ORIGINATOR INDICATOR

CTSW108	**Incoming AMQP message with known originator indicator**
Test criteria	This test is successful, if the IUT correctly converts an AMQP message containing a known originator into an AMHS message (IPM).
Scenario description	From the AMQP test interface send an AMQP message to the IUT. The AMQP message shall contain an originator indicator with an 8 letters address which is known in the IUT and that can be converted. Check the IPM that the AMHS Test Tool receives. Verify that the IUT has converted the message correctly accordingly: • *amhs_originator* shall be converted into a MF-address following conversion of AFTN/AMHS addresses. • Also, it shall be conveyed as the value of the *originator* in *this-IPM* element of the IPM heading field and as the *originator-name* in the envelope of the AMHS message.
Ref Doc 047	4.5.2.12 a) amhs_originator 4.5.3.2 originator heading field 4.5.3.5 formal-name 4.5.3.6 element user in this-IPM heading field 4.5.4.6 originator-name
Test class	Normal communications (N)

CTSW109 – INCOMING AMQP MESSAGE WITH UNKNOWN ORIGINATOR INDICATOR

CTSW109	**Incoming AMQP message with unknown originator indicator**
Test criteria	This test is successful, if the IUT logs and reports the AMQP message to the Control Position, if during the conversion process the translation of the originator indicator fails. In this case, the default originator shall be used.
Scenario description	From the AMQP test interface send an AMQP message to the IUT. The AMQP message shall contain an originator indicator which is unknown in the IUT and therefore it cannot be converted. Verify that the IUT converts and sends the message to the AMHS Test Tool using the default originator specified in the AMHS/SWIM Gateway's configuration parameters. Check that this situation has been logged and reported to the Control Position.
Ref Doc 047	4.5.2.12 amhs_originator (default originator)
Test class	Normal communications (N)

CTSW110 – REJECT AN AMQP MESSAGE WITH UNSUPPORTED CONTENT-TYPE

CTSW110	**Reject an AMQP message with unsupported content-type**
Test criteria	This test is successful, if the IUT's gateway component rejects an incoming message if it does not meet with the following assertions: • content-type contains the value *"<application/octet-stream>",* the *data* element is present, and the *amqp-value* element is not present. • content-type contains the value *"<text/plain; charset="utf-8">"*, the *amqp-value* element is present, and the *data* element is not present
Scenario description	From the AMQP test interface send a sequence of messages to the IUT addressed to an AMHS user recipient. The messages shall have different values for the content-type: • The 1st message shall contain a *content-type* value "*<application/octet-stream>", while amqp-value* and *data* shall be empty. • The 2nd message shall contain a *content-type* value *"<text/plain; charset="utf-8">", amqp-value* shall be empty, and *data* shall contain binary data. • The 3rd message shall contain a *content-type* value "*<application/octet-stream>", amqp-value* shall be empty, and *data* shall contain binary data. • The 4th message shall contain a *content-type* value *"<text/plain; charset="utf-8">", amqp-value* shall be present and *data* shall be empty. • Optionally, the 5th message shall contain a *content-type* value *"<text/plain; charset="utf-16">"* • Optionally, the 6th message shall contain a *content-type* value *"<text/plain; charset="utf-8">", amqp-value* shall be present and *data* shall contain binary data. Verify that the IUT accepts and converts the 3rd and 4th messages, but rejects 1st, 2nd, 5th and 6th messages. Check that this situation has been logged and reported to Control Position.
Ref Doc 047	4.5.1.6 content-type
Test class	Erroneous AMQP messages (E1)

CTSW111 – CHECK IF THE SIZE IN BYTES OF THE PAYLOAD OF AN AMQP MESSAGE EXCEEDS THE MAXIMUM CONFIGURED

CTSW111	**Check if the size in bytes of the payload of an AMQP message exceeds the maximum configured**
Test criteria	This test is successful, if the IUT rejects a received AMQP message if the size of the payload in bytes of the message exceeds the maximum value specified in the AMHS/SWIM Gateway's configuration parameter "Maximum message data size."
Scenario description	From the AMQP test interface send three AMQP messages to the IUT transfer port. The messages shall address an AMHS user. a) Send one AMQP message, the size of the *amqp-value* shall not exceed the maximum value configured. b) Send one more AMQP message, the size of the payload in bytes for *data* shall not exceed the maximum value configured. Verify that these two messages are received at the AMHS test interface. c) Send one AMQP message, the size of the *amqp-value* shall exceed the maximum value configured. d) Send one more AMQP message, the size of the payload in bytes for *data* shall exceed the maximum value configured. Verify that for the last two messages the IUT does not convert the AMQP messages into AMHS format, but rejects the messages, logs, and reports the situation to the Control Position.
Ref Doc 047	4.5.1.7 Maximum message data size
Test class	Procedural errors (E2) Normal communications (N)

CTSW112 – CHECK IF AN AMQP MESSAGE ADDRESSES MORE AMHS USERS THAN THE MAXIMUM CONFIGURED

CTSW112	**Check if an AMQP message addresses more AMHS users than the maximum configured**
Test criteria	This test is successful, if the IUT rejects a received AMQP message addressing a number of message recipients, which the ITCU is responsible for conveyance, that exceeds the maximum value defined in the AMHS/SWIM Gateway's configuration parameter "Maximum message number of recipients."
Scenario description	From the AMQP test interface send two AMQP messages to the IUT. a) Send one AMQP message with 512 AMHS recipients. The "Maximum message number of recipients" parameter is set to 512. Verify that this message is received at the AMHS test interface. b) Send one more AMQP message with 513 recipients. The "Maximum message number of recipients" parameter is set to 512. Verify that the IUT does not convert the AMQP message into AMHS format, but rejects the message, logs, and reports the situation to the Control Position.
Ref Doc 047	4.5.1.8 Maximum message number of recipients
Test class	Procedural errors (E2) Normal communications (N)

CTSW113 – INCOMING RECEIPT AND NON-RECEIPT NOTIFICATIONS (RN AND NRN)

CTSW113	Incoming receipt and non-receipt notification (RN and NRN)
Test criteria	This test is successful, if the IUT reports to its control position and stores the message, if it receives an IPN containing a NRN or RN addressed to an AMQP consumer.
Scenario description	From the AMQP test interface send two AMQP messages with priority “6” to the IUT. These messages shall address an AMQP consumer and be converted by the IUT into AMHS format and sent to the AMHS Test Tool. Remove Message 1 from the AMHS Test Tool For the first one, upon reception of the AMHS message, the AMHS Test Tool returns an NRN. Verify that the IUT behaves as specified in section 4.4.7.3 of EUR Doc 047, i.e.: • logs the situation and reports to Control Position, and • stores the message for appropriate processing at the Control Position. For the second one, upon reception of the AMHS message the AMHS Test Tool returns a RN. Check the element *notification-requests* in the AMHS message. The values “RN” and “NRN” shall be taken simultaneously by this element. Verify that the IUT behaves the same way as mentioned before and: • logs the situation and reports to Control Position, and • stores the message for appropriate processing at the Control Position.
Ref Doc 047	4.5.3.4 Values of notification-request element: “rn” and “nrn” 4.4.7.3 RN valid
Test class	Normal communications (N)

CTSW114 – INCOMING NON-DELIVERY REPORTS (NDR)

CTSW114	**Incoming non-delivery reports (l)**
Test criteria	This test is successful, if the IUT logs and reports to its control position, if it receives an NDR.
Scenario description	From the AMQP test interface send an AMQP message to the IUT. The message shall address an AMHS user and be converted by the IUT into AMHS format. This message shall be queued in the AMHS Test Tool. After being queued, it shall be deleted(sending report to originator). Verify that an NDR is received by the IUT with the following elements: • "unable-to-transfer" for the *non-delivery-reason-code*, • empty field for the *non-delivery-diagnostic-code* , Check that the IUT logs and reports the situation to the Control Position.
Ref Doc 047	4.4.1.3 Report reception
Test class	Normal communications (N)

CTSW115 – INCOMING AMQP MESSAGES WITH AMQP-VALUE AND DIFFERENT AMHS_BODYPART_TYPE AND AMHS_CONTENT_ENCODING VALUES

CTSW115	Incoming AMQP text messages with amqp-value and different amhs_bodypart_type and amhs_content_encoding values
Test criteria	This test is successful, if the IUT's gateway component converts received AMQP amqp-value and messages with different *amhs_bodypart_type and amhs_content_encoding* values to AMHS messages (IPM) with appropriate body-part type and encoding.
Scenario description	From the SWIM Test Tool send a sequence of AMQP messages (IPMs) to the IUT addressed to an AMHS user reachable via the IUT's gateway. All messages shall contain *<text/plain; charset=\"utf-8\">* content type and: • The first AMQP message shall have for *amhs_bodypart_type* set to "ia5-text", *amhs_content_encoding* set to *"*IA5*"* and *amqp-value* set to "Lorem ipsum". • The second AMQP message shall have for *amhs_bodypart_type* set to "ia5_text_body_part", *amhs_content_encoding* set to *"*IA5*"* and *amqp-value* set to "Lorem ipsum i5bpt". • The third AMQP message shall have for *amhs_bodypart_type* set to "general-text-body-part", *amhs-content-encoding* set to *"*ISO-646*"* and *amqp-value* set to "Lorem ipsum 646". • The fourth AMQP message shall have for *amhs_bodypart_type* set to "general-text-body-part", *amhs_content_encoding* set to *"*ISO- 8859-1*"* and *amqp-value* set to "Lorem ipsum 8859". Check messages received at the AMHS interface. Check the correct format of the AMHS messages according to the Table 10 of EUR Doc 047. Verify that the *ATS-message-text* of the AMHS messages is equal to the text set in the *amqp-value* for each message. Verify that the original-encoded-information-types element is generated as detailed in section 4.5.4.7 of EUR Doc 047.
Ref Doc 047	4.5.2.4 amhs_bodypart_type and amhs_content_encoding 4.5.2.5 amhs_bodypart_type 4.5.2.14 a) amqp-value 4.5.4.7 original-encoded-information-types Table 10. Mapping of AMQP Message (AMHS aware) Table 11. Mapping of AMQP Message (AMHS unaware)
Test class	Erroneous AMQP messages (E1)

CTSW116 – INCOMING AMQP BINARY MESSAGE WITH FTBP ATTRIBUTES

CTSW116	Incoming AMQP binary message with FTBP attributes
Test criteria	This test is successful, if the IUT's gateway component converts a received AMQP binary message to an AMHS message (IPM) with a FTBP and appropriate attributes.
Scenario description	From the SWIM Test Tool send a sequence of AMQP messages (IPMs) to the IUT addressed to an AMHS user reachable via the IUT's gateway. • The first message shall contain <*application/octet-stream*> content-type and an empty *amqp-value*. Also, *data* element shall contain a binary file and the application properties amhs_ftbp_file_name, amhs_ftbp_object_size and amhs_ftbp_last_mod. • The second message shall contain the same elements, and additionally the application property swim_compression shall contain value "gzip". In this case, the binary payload shall be a compressed content. Check the messages received at AMHS interface. Check the correct format of the AMHS message, which shall contain a file-transfer-body-part. Verify that AMHS properties related to FTBP are correct: • *incomplete-pathname* element of the file transfer parameters of the AMHS message shall be conveyed using *amhs_ftbp_file_name* element of AMQP message. • *actual-values* element of the IPM file transfer parameters of the AMHS message shall be conveyed using *amhs_ftbp_object_size* element of AMQP message. • *date-and-time-of-last-modification* element of the IPM file transfer parameters of the AMHS message shall be conveyed using *amhs_ftbp_last_mod* element of AMQP message if present. In the second message, verify that the payload has been uncompressed. Verify that the *ATS-message-text* element is not present in the generated AMHS messages (4.5.2.14 b)
Ref Doc 047	4.5.2.6 amhs_ftbp_file_name 4.5.2.7 amhs_ftbp_object_size 4.5.2.8 amhs_ftbp_last_mod 4.5.2.13 data 4.5.2.14 b) amqp-value
Test class	Normal communications (N)